

“The Essentials of Computer Organization and Architecture”

Unit 5 – Exercises and Solutions

EAT1 Consider a system with a 3-level memory hierarchy, with average access times of 20ns, 200ns e 2ms. For level 1 and level 2, the average hit rates are 90% e 95%. Compute the average access time of this hierarchy.

EAT2 A system with a 4-level memory hierarchy has average access times of 1ns, 10ns, 100ns and 1 μ s. For the first 3 levels, the average hit rates are 90%, 94% and 97%. Compute the average access time of this hierarchy.

EAT3 a) Deduce the EAT expression for a 5-level memory hierarchy. b) Apply it reusing the times and probabilities of exercise EAT2, and considering a hit-rate in level 4 of 99% and access time of 1ms in level 5.

MP1 Consider a 2M x 32 bit memory. Answer the following questions:



- a) What is the storage capacity of this memory (in bytes) ?
- b) How many bits have the addresses of this memory, if the memory is byte-addressable?
- c) c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 32 bits)

MP2 Consider a 4M x 16 bit memory. Answer the following questions:



- a) What is the storage capacity of this memory (in bytes) ?**
- b) How many bits have the addresses of this memory, if the memory is byte-addressable?**
- c) How many bits have the addresses of this memory, if the memory is word-addressable? (note: consider the word-size as 16 bits)**

MP3 Consider a 1M x 8 bit memory.

Answer the following questions:

- a) How many bits have the addresses of this memory, if the memory is byte-addressable?**
- b) How many bits have the addresses of this memory, if the memory is word-addressable?**

MP4 Consider a memory of 2^{20} bytes. Answer the following questions:



- a) What is the lowest and highest address if this memory is byte-addressable?**
- b) What is the lowest and highest address if this memory is word-addressable?
(note: consider the word size as 16 bits)**
- c) What is the lowest and highest address if this memory is word-addressable?
(note: consider the word size as 32 bits)**

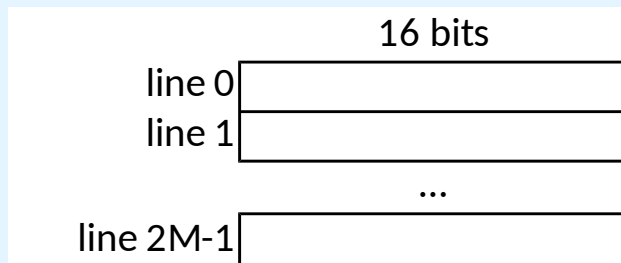
MP5 A $2M \times 16$ main memory is built using $256K \times 8$ chips and memory is word-addressable. Provide:

- a) The number of chips necessary per each memory word.**
- b) The total number of modules of the memory.**
- c) The total number of chips of the memory.**
- d) The number of bits of the memory addresses.**
- e) The number of bits used to select the module.**
- f) The number of bits used to define the offset (line) within a module.**
- g) The module and the offset (line) within that module, for the memory address 14, under g1) High-Order and g3) Low-Order interleaving**
- h) The overall storage capacity (in bytes) of the memory.**

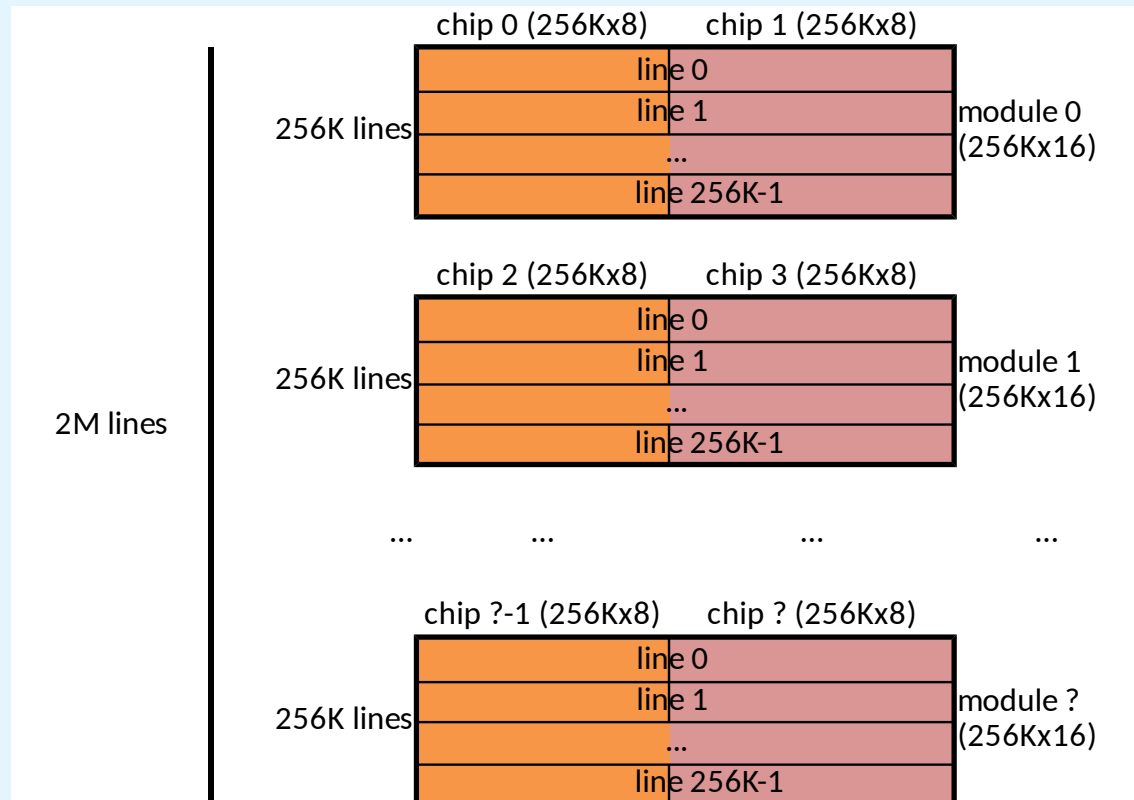
MP5 A 2M x 16 main memory is built using 256K × 8 chips and memory is word-addressable.

Based on the data given, one can build the following representations:

Abstract View of Memory



Physical View of Memory



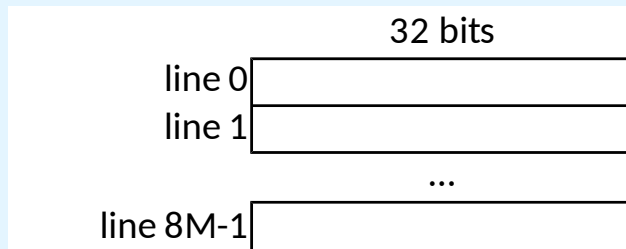
MP6 A 8M x 32 main memory is built using 512K × 16 chips and memory is word-addressable. Provide:

- a) The number of chips necessary per each memory word.**
- b) The total number of modules of the memory.**
- c) The total number of chips of the memory.**
- d) The number of bits of the memory addresses.**
- e) The number of bits used to select the module.**
- f) The number of bits used to define the offset (line) within a module.**
- g) The module and the offset (line) within that module, for the memory address 41, under g1) High-Order and g3) Low-Order interleaving**
- h) The overall storage capacity (in bytes) of the memory.**

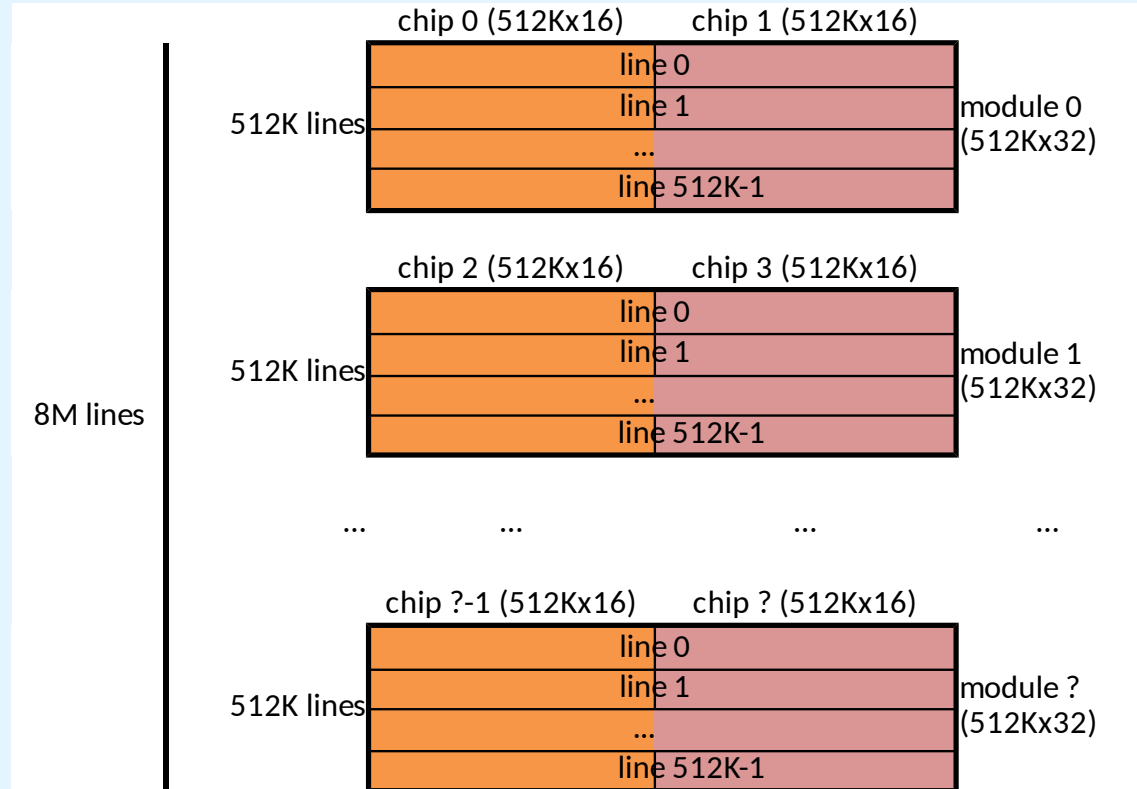
MP6 A 8M x 32 main memory is built using 512K x 16 chips and memory is word-addressable.

Based on the data given, one can build the following representations:

Abstract View of Memory



Physical View of Memory



MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

a) Provide the number of chips necessary per each memory word:

{?} chips

b) Provide the total number of modules of the memory:

{?} modules

c) Provide the total number of chips of the memory:

{?} chips

d) Provide the number of bits of the memory addresses:

{?} bits

e) Provide the number of bits used to select the module:

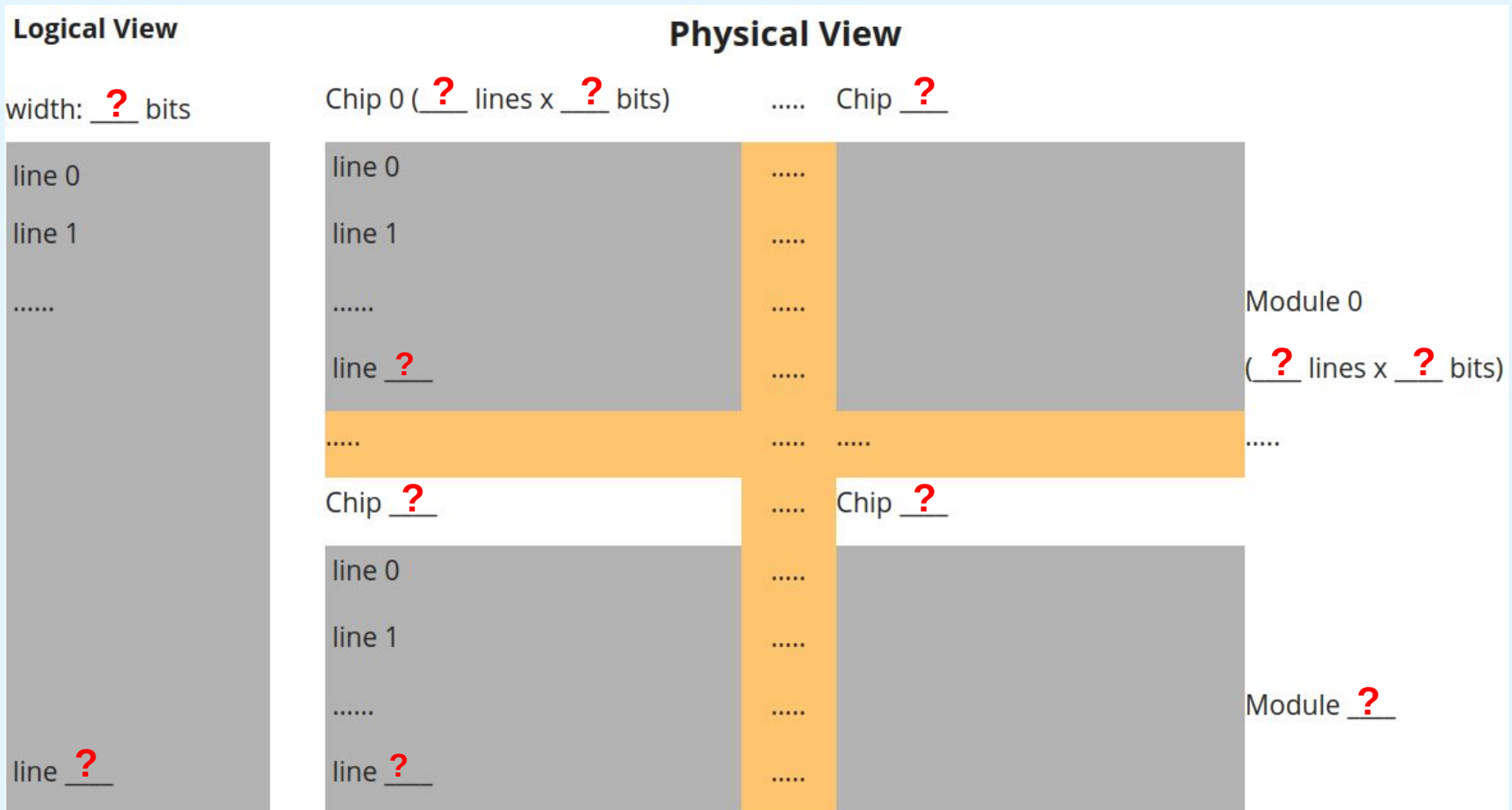
{?} bits

f) Provide the number of bits used to define the offset (line) within a module:

{?} bits

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

g) Complete the representation bellow of the logical & physical view of memory:



MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 3333, considering the following alternatives for the mapping of a memory address into its physical components:

h1) High-Order Interleaving:

module: {?}

offset: {?}

h2) Low-Order Interleaving:

module: {?}

offset: {?}

MP7 A memory of 32G x 64 bits is built with 1G x 32 chips and is word-addressable (64 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

{?} bytes

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

a) Provide the number of chips necessary per each memory word:

{?} chips

b) Provide the total number of modules of the memory:

{?} modules

c) Provide the total number of chips of the memory:

{?} chips

d) Provide the number of bits of the memory addresses:

{?} bits

e) Provide the number of bits used to select the module:

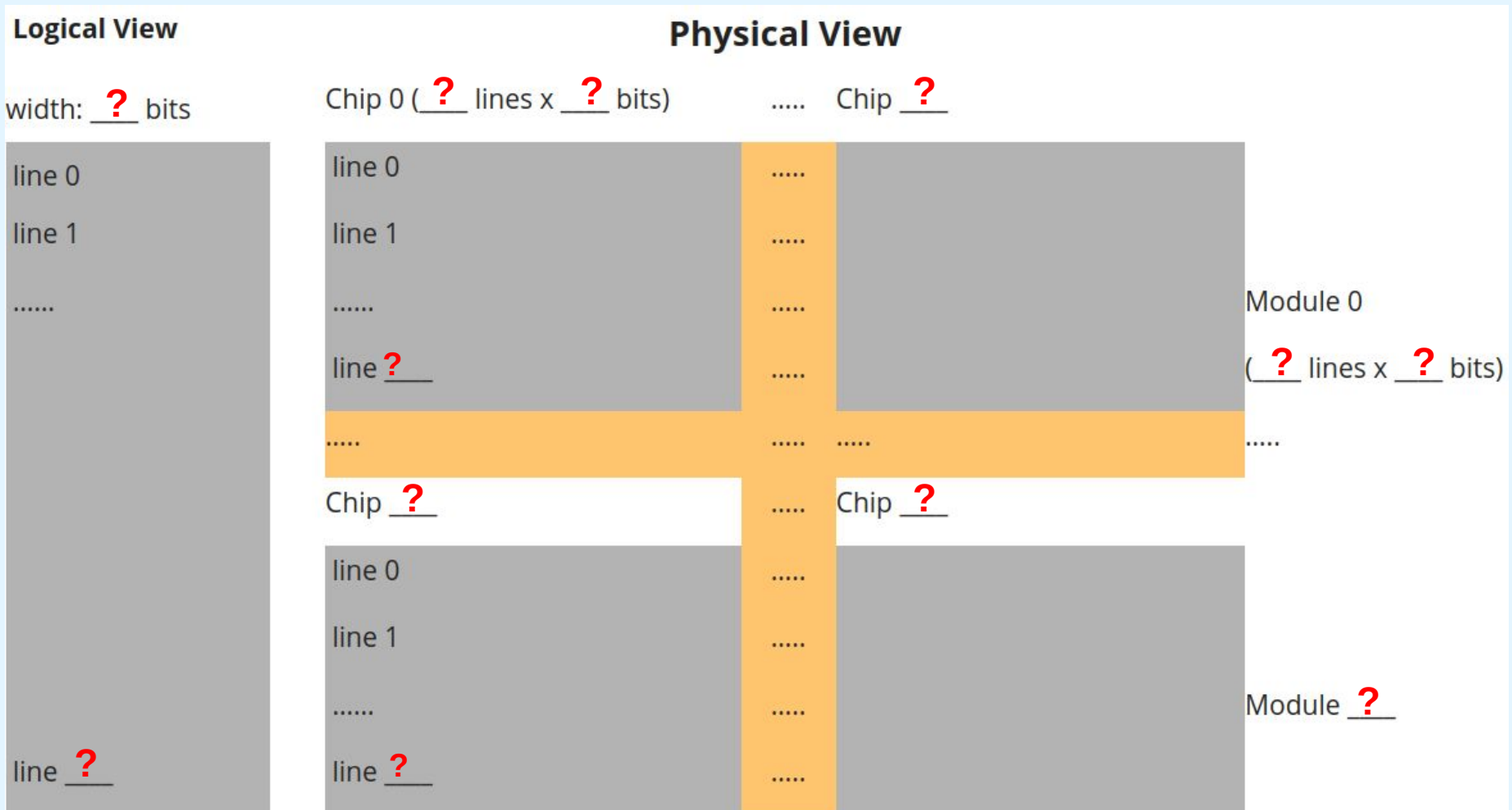
{?} bits

f) Provide the number of bits used to define the offset (line) within a module:

{?} bits

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

g) Complete the representation bellow of the logical & physical view of memory:



MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

h) Find and present the module and the offset (line) within that module, for the memory address 8888888, considering the following alternatives for the mapping of a memory address into its physical components:

h1) High-Order Interleaving:

module: {?}

offset: {?}

h2) Low-Order Interleaving:

module: {?}

offset: {?}

MP8 A memory of 16G x 32 bits is built with 256M x 8 chips and is word-addressable (32 bits word size). Answer the following questions.

i) Provide the overall storage capacity (in bytes) of the memory.

{?} bytes

MP9 Given a memory of 2048 bytes consisting of several 64 Byte x 8 chips, and assuming byte-addressable memory, which of the following seven alternatives uses the address bits correctly ?

- a.

2 bits for chip select	8 bits for address on chip
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- b.

16 bits for chip select	48 bits for address on chip
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- c.

6 bits for chip select	5 bits for address on chip
------------------------	----------------------------
- d.

1 bit for chip select	5 bits for address on chip
-----------------------	----------------------------
- e.

5 bits for chip select	6 bits for address on chip
------------------------	----------------------------
- f.

4 bits for chip select	6 bits for address on chip
------------------------	----------------------------
- g.

8 bits for chip select	56 bits for address on chip
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CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

- a) How many blocks of memory are there ?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $0DB63_{16}$ map ?**
- d) To which memory block and word will the memory address $0DB63_{16}$ map ?**

Note: in this and in the following exercises on caches, “memory” refers to the “main memory”, also sometimes referred to as “RAM”.

CACHE1 A computer has a main memory of 2^{20} words and a direct mapped cache of 32 blocks, where each cache block contains 16 words.

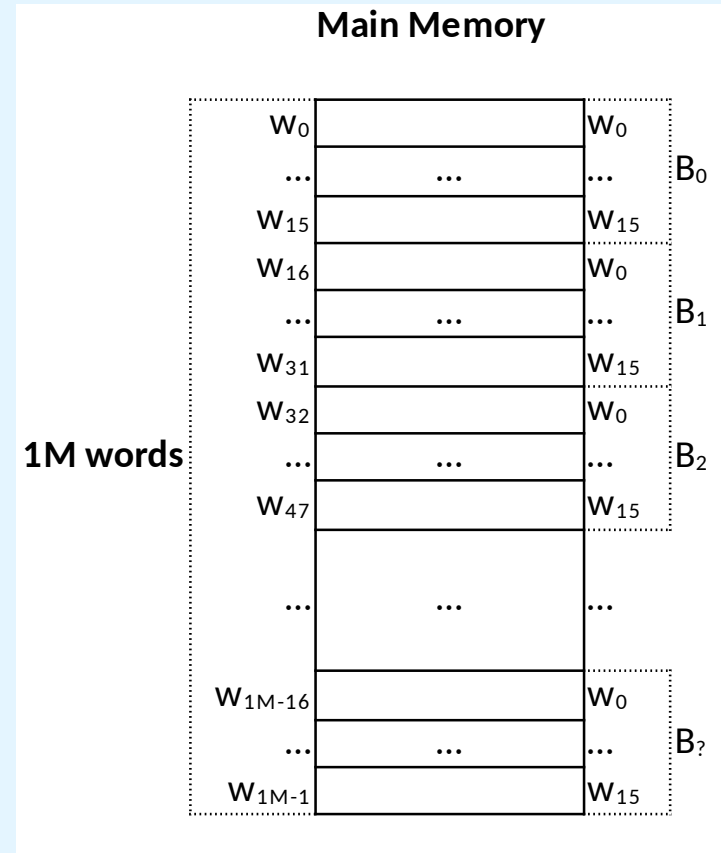
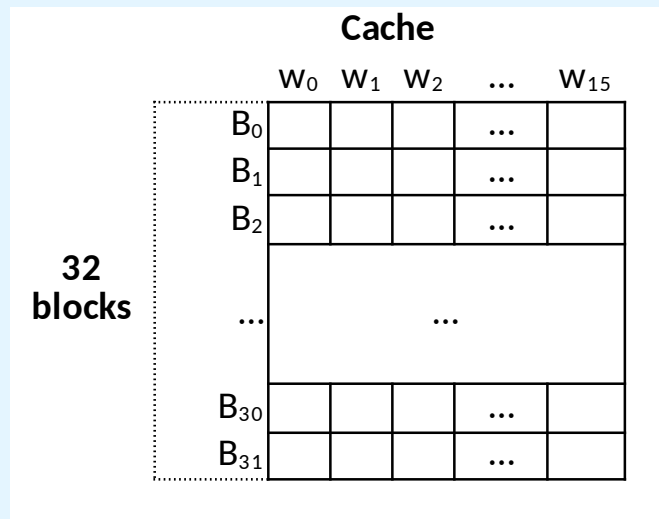
Based on the data given, one can build the following representations:

Problem Data:

#WORDS(memory) = 1M = 2^{20} words

#BLOCKS(cache) = 32 = 2^5 blocks

#WORDS(cache block) = 16 = 2^4 words



CACHE2 A computer has a main memory of 4G words and a direct mapped cache of 1K blocks, where each cache block contains 32 words.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block the memory address $000063FA_{16}$ maps ?**
- d) To which memory block, and block word, the memory reference $000063FA_{16}$ maps ?**

Homework

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

- a) What is the format of a memory address as seen by the cache ?**
- b) Compute the hit ratio for a program that loops 2 times from locations 0 to 67 in memory. The cache is initially empty.**
- c) Assume access time for the cache is 22ns and the time required to fill a cache slot from main memory is 300ns. Compute the effective access time in the b) conditions.**

CACHE3/EAT4 A computer has a main memory of 4K blocks (of 8 words per block) and a direct-mapped cache of 8 blocks.

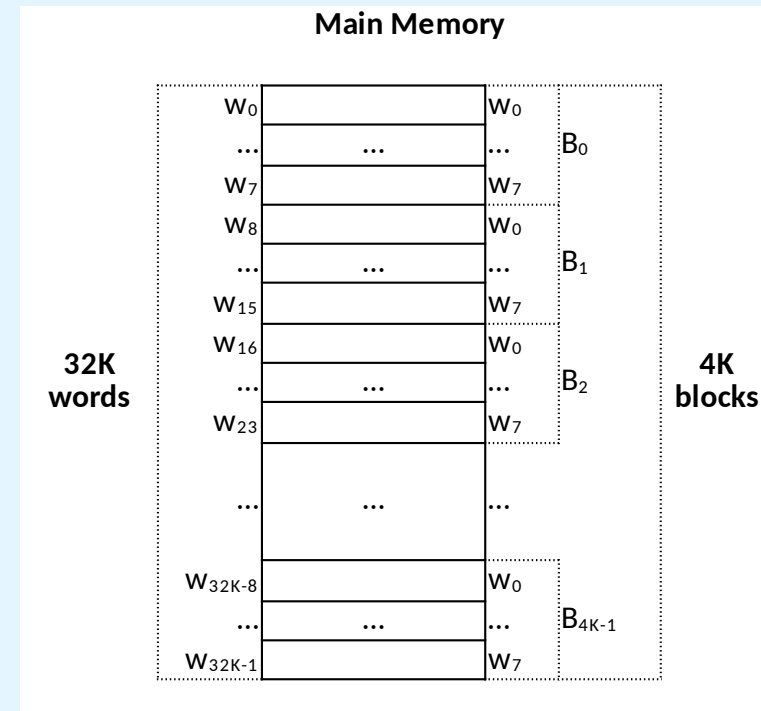
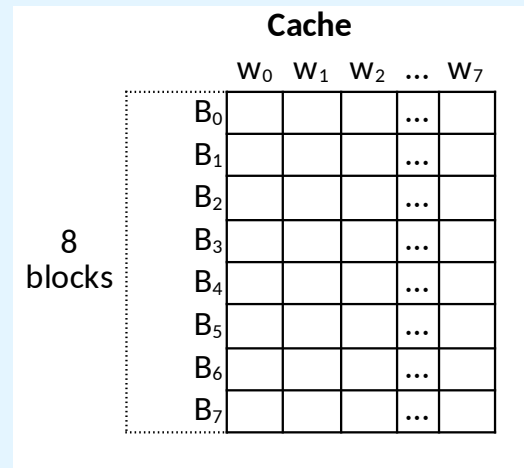
Based on the data given, one can build the following representations:

Problem Data:

#BLOCKS(memory) = 4K blocks = $2^2 \times 2^{10} = 2^{12}$ blocks

#WORDS(memory block) = 8 = 2^3 words

#BLOCKS(cache) = 8 blocks = 2^3 blocks



Consequences:

#WORDS(cache block) = #WORDS(memory block) = 8 words

#WORDS(memory) = #BLOCKS(memory) x #WORDS(memory block)
 $= 2^{12} \times 2^3 = 2^{15} = 32K$ words

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $F89C_{16}$ map ?**
- d) To which memory block, and block word, will the memory address $F89C_{16}$ map ?**

CACHE4 A computer has a main memory of 2^{16} words and a fully associative cache of 64 blocks, with 32 words per cache block.

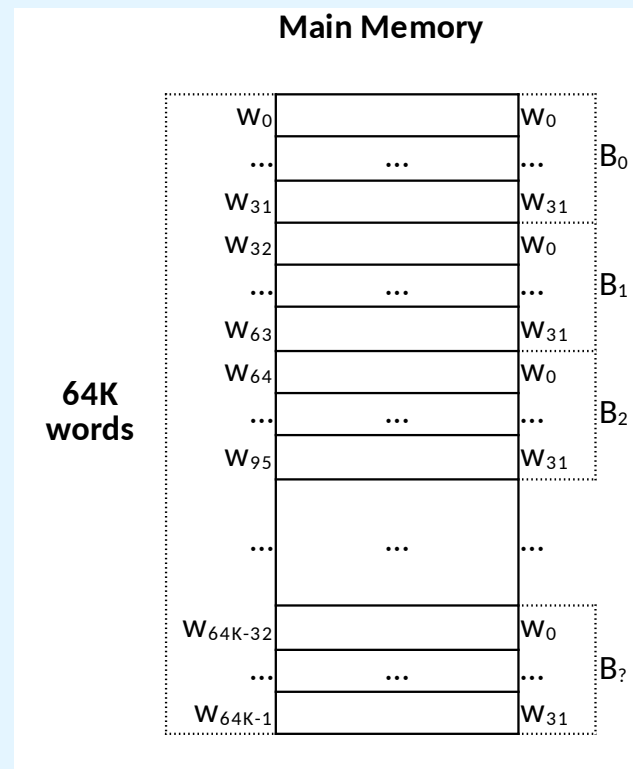
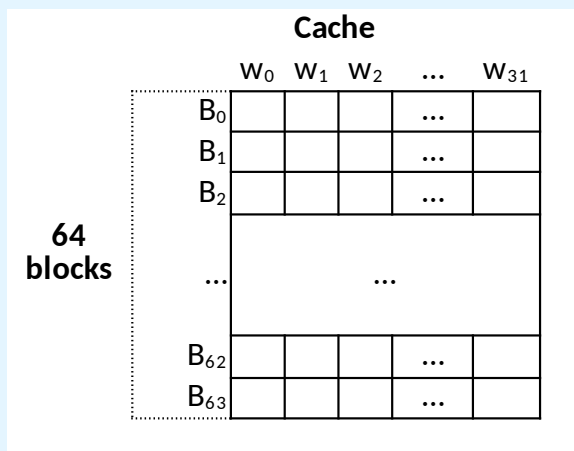
Based on the data given, one can build the following representations:

Problem Data:

#WORDS(memory) = 64K = 2^{16} words

#BLOCKS(cache) = 64 = 2^6 blocks

#WORDS(cache block) = 32 = 2^5 words



CACHE5 A computer has a main memory of 16M words and a fully associative cache of 128 blocks, with 64 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache block will the memory address $01D872_{16}$ map ?**
- d) To which memory block, and block word, will the memory address $01D872_{16}$ map ?**

Homework

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

- a) How many blocks of memory are there?**
- b) What is the format of a memory address as seen by the cache ?**
- c) To which cache set, and block of that set, will the memory address $0DB630D_{16}$ map?**
- d) To which memory block, and block word, will the memory address $0DB630D_{16}$ map ?**

CACHE6 A computer has a main memory of 128M words and a 2-way set associative cache of 32K blocks, with 64 words per cache block.

Based on the data given, one can build the following representations:

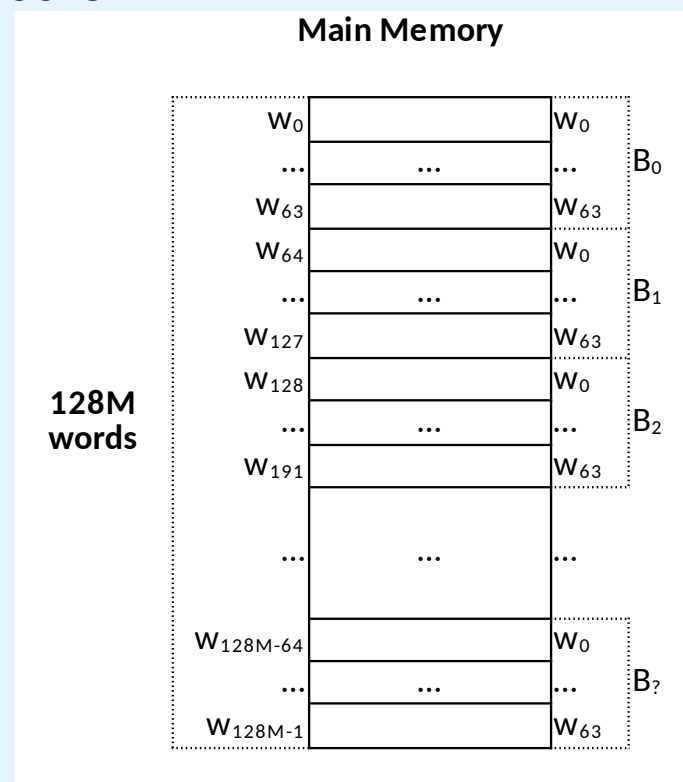
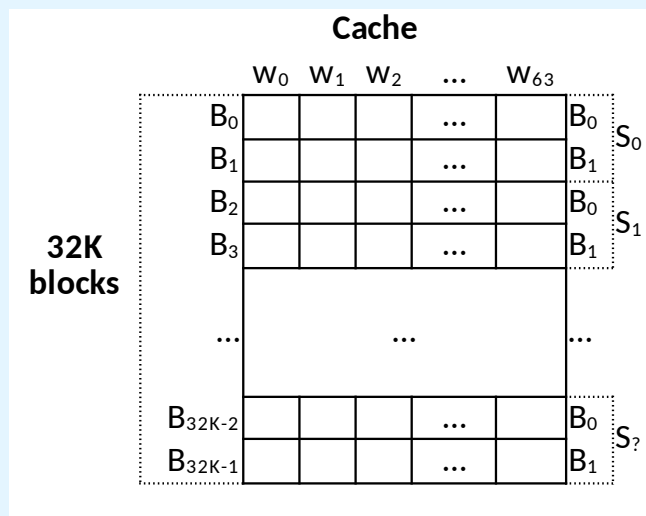
Problem Data:

#WORDS (memory) = 128M = $2^7 \times 2^{20} = 2^{27}$ words

N = number of ways = number of cache blocks per cache set = 2

#BLOCKS(cache) = 32K = $2^5 \times 2^{10} = 2^{15}$ blocks

#WORDS(cache block) = 64 = 2^6 words



CACHE7 A computer has a byte-addressable memory with 24-bit addresses, and a 64KB cache with 32 bytes blocks.

Show the format of a memory address as seen by the cache if the cache is:

- a) direct mapped**
- b) (fully) associative**
- c) 4-way set associative**

Homework

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

- a) What is the format of a memory address as seen by the cache ?**
- b) Compute the hit ratio for a program that loops 3 times from locations 8 to 23, and 40 to 47, in memory.**

CACHE8/EAT5 A computer has a main memory of 2K blocks, of 8 words each, and a 2-way set associative cache, with 4 sets of blocks.

Based on the data given, one can build the following representations:

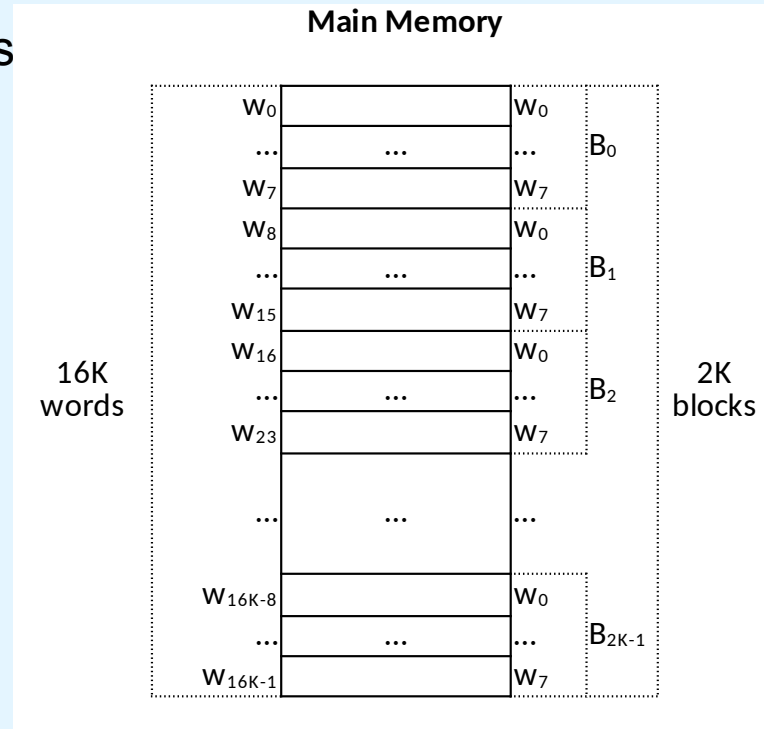
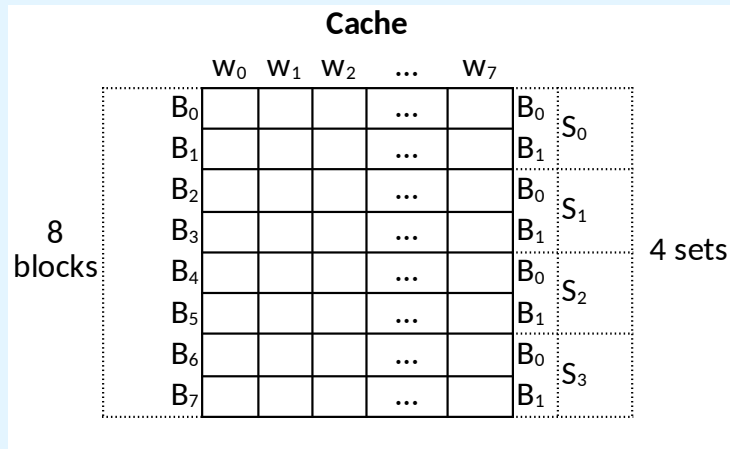
Problem Data:

#BLOCKS(memory) = $2K = 2^1 \times 2^{10} = 2^{11}$ blocks

#WORDS(memory block) = $8 = 2^3$ words

N = num. of ways = num. of blocks per set = 2

#SETS(cache) = $4 = 2^2$ sets



Consequences:

#BLOCKS(cache) = #SETS(cache) $\times$ N = $4 \times 2 = 8 = 2^3$ blocks

#WORDS(memory) = #BLOCKS(memory) $\times$ #WORDS(memory block)

= $2^{11} \times 2^3 = 2^{14} = 16K$ words